

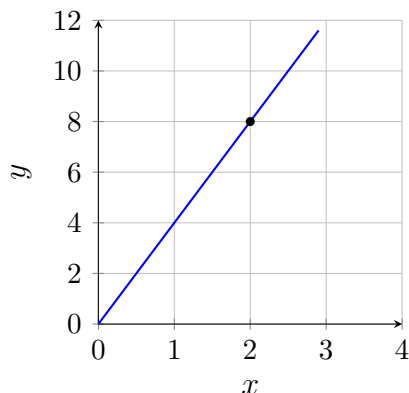
Proportional Relationships: Four Representations

Moving between graphs, tables, equations, and real-world contexts

Directions:

- Every proportional relationship can appear four ways: a **table**, a **graph** (straight line through the origin), an **equation** $y = kx$, and a **story**. Each problem asks you to move from one to another.
- After the warm-up the representations mix, so read each problem carefully before choosing a method. The last problem combines them all.

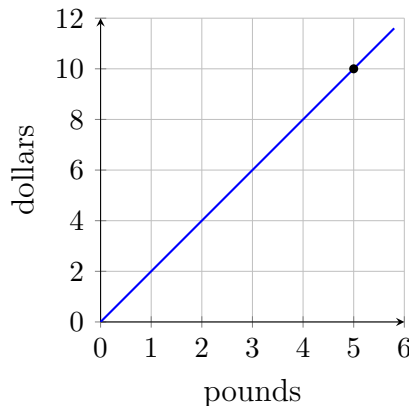
1. Warm-up. The graph shows a proportional relationship. Write its equation in the form $y = kx$.



Work space:

Answer: _____

2. Warm-up. The graph shows the cost y , in dollars, of x pounds of apples. Find the unit rate, in dollars per pound.

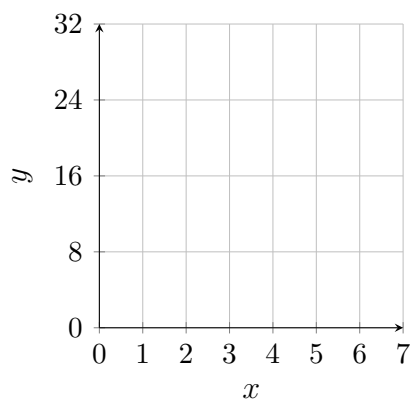


Work space:

Answer: _____

3. The table shows a proportional relationship.

x	2	4	6
y	9	18	27

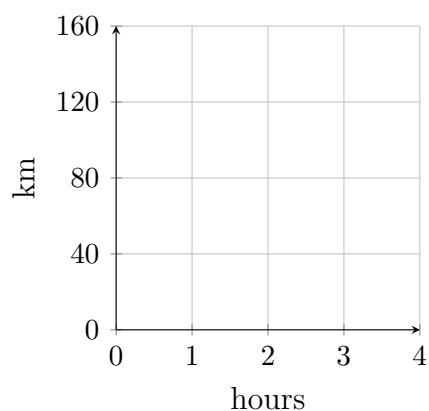


(a) Write the equation: _____

(b) Use the equation to find y when $x = 10$: _____

You may plot the points to check your equation.

4. A tour bus travels at a steady 40 kilometers per hour.

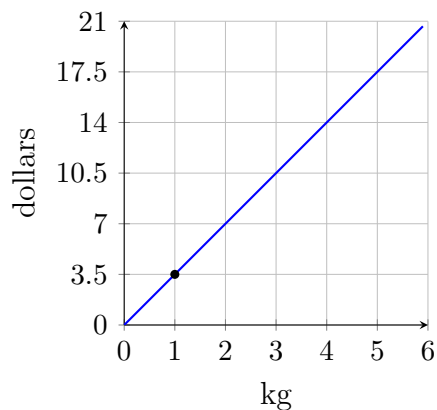


(a) Complete the table of distances:

hours	km
1	_____
2	_____
3	_____

(b) Plot the three points on the grid and join them with a line through the origin.

5. The equation $y = 3.5x$ gives the cost y , in dollars, of x kilograms of rice. Its graph is shown.

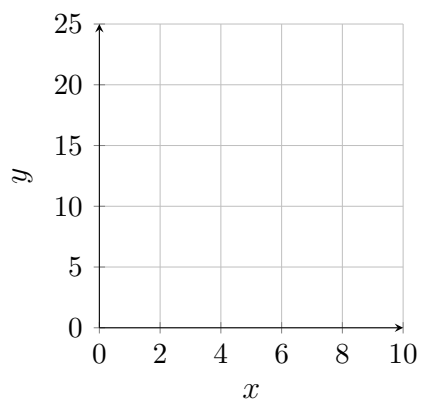


(a) Cost, in dollars, of 6 kg: _____

(b) The point $(1, 3.5)$ is marked on the graph. Describe what it tells you about the price of rice.

6. The table shows a proportional relationship.

x	3	5
y	7.5	12.5

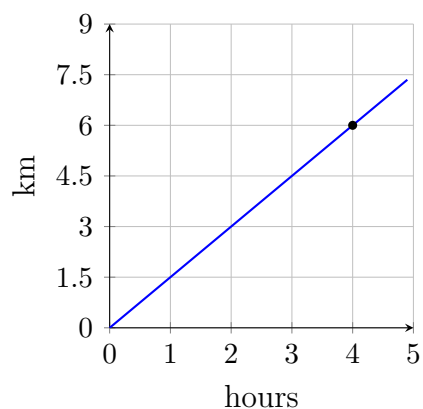


(a) Write the equation: _____

(b) Find x when $y = 20$: _____

You may plot the points to check your equation.

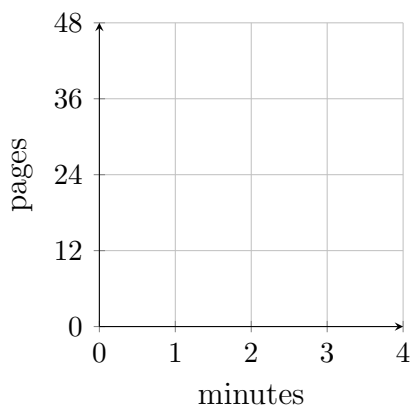
7. The graph shows how far a hiker walks: y kilometers in x hours.



(a) Write the equation: _____

(b) How far does the hiker walk in 10 hours, in km? _____

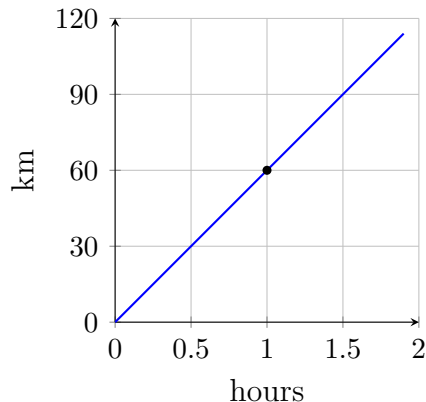
8. A school printer prints 12 pages per minute.



(a) Sketch the graph of pages printed against minutes on the grid.

(b) Use your graph or an equation to find how many pages print in 2.5 minutes: _____

9. A train's distance is given by the equation $y = 60x$, where y is in kilometers and x is in hours. Its graph is shown.

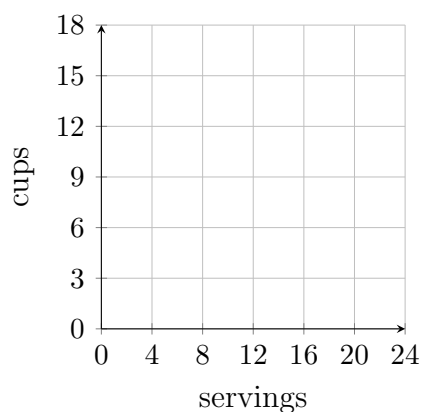


(a) How many kilometers does the train cover in 1.5 hours? _____

(b) In your own words, say what the number 60 in the equation tells you about this train.

10. A soup recipe's table relates servings (x) to cups of broth (y).

servings (x)	4	8
cups of broth (y)	3	6

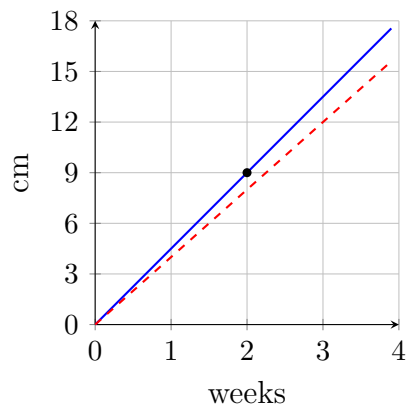


(a) Write the equation: _____

(b) How many servings can be made from 15 cups of broth? _____

You may plot the points to check your equation.

11. Plant A's growth is graphed below (solid line): it passes through $(2, 9)$, with y in centimeters and x in weeks. Plant B grows according to the equation $y = 4x$. How much more does Plant A grow *per week* than Plant B, in centimeters?

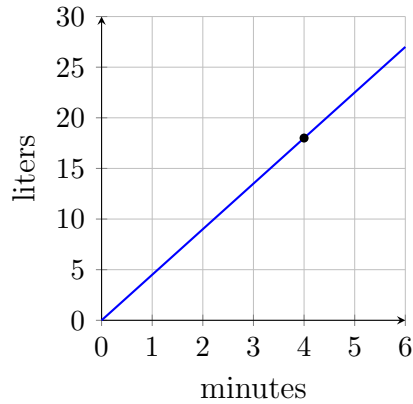


Work space:

Answer: _____

12. Challenge. Tank A fills according to the graph below: it passes through $(4, 18)$, with y liters after x minutes. Tank B fills according to this table:

minutes (x)	2	6
liters (y)	10	30



(a) Equation for Tank A: _____

(b) Compare the fill rates with $<$, $>$, or $=$:

Tank A's rate _____ Tank B's rate

(c) After 8 minutes, how many *more* liters are in the faster tank? _____